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These solutions are not the only "perfect" or "10/10" solutions to be followed as a model. They are examples of ways to solve these problems. There are many many alternative ways to solve these problems. We hope that these example solutions help you check your work.

Problems

Q1. Consider the set of polynomials with real coefficients $\mathbf{P}(\mathbf{R})$. Formally,

$$\mathbf{P}(\mathbf{R}) = \bigcup_{n=0}^{\infty} \mathbf{P}_n(\mathbf{R}).$$

Prove that $\mathbf{P}(\mathbf{R})$ is NOT finite dimensional.

Solution: Note that the monomials $\{1, x, x^2, \ldots, x^n, \ldots\}$ form an infinite linearly independent set. If $\mathbf{P}(\mathbf{R})$ where spanned by finitely many vectors, say $\{p_1(x), \ldots, p_n(x)\}$, and k where the maximum between the degrees of the $p_i(x)$'s, then $x^{k+1} \in \operatorname{span}\{1, \ldots, x^k\}$, which is a contradiction.

Q2. Let $f(x)$ be a polynomial in $\mathbf{P}_n(\mathbf{R})$ with degree n. For example, $x^2 + 1 \in \mathbf{P}_n(\mathbf{R})$ has degree two.

Prove: For any polynomial $g(x) \in \mathbf{P}_n(\mathbf{R})$ there are constants $c_1, c_2, \ldots, c_{n+1} \in \mathbf{R}$ such that:

$$g(x) = c_1 f(x) + c_2 f'(x) + \cdots + c_{n+1} f^{(n)}(x)$$

where $f^{(k)}(x)$ is the kth derivative of $f(x)$.

Solution: Let $f(x) = a_0 + a_1 x + \cdots + a_n x^n$. Since $f(x)$ has degree n, $a_n \neq 0$. Let now $1 \leq k \leq n$, and note that the leading coefficient of $f^{(k)}(x)$, meaning, the coefficient multiplying the highest power of x in $f^{(k)}(x)$ (which is x^{n-k}) is $n(n-1)(n-2)\ldots(n-(k-1))a_n$, which is also non-zero. This implies that the polynomial $f^{(k)}(x)$ has degree $n-k$. Thus, the family $\alpha := \{f(x), f'(x) \ldots, f^{(n)}(x)\}$ is a set of $n+1$ linearly independent vectors in $\mathbf{P}_n(\mathbf{R})$, so it has to be a basis. In particular, every polynomial in $\mathbf{P}_n(\mathbf{R})$ is a linear combination of elements of α, as desired.

Q3. The vectors $\{(1,2,3), (2,1,0)\}$ are linearly independent in $\mathbb{R}^3$.
Find a vector $\mathbf{v}$ such that $\{(1,2,3), (2,1,0), \mathbf{v}\}$ is a basis of $\mathbb{R}^3$.

Solution: Note that every element of the span of $\{(1,2,3), (2,1,0)\}$ is of the form

$$\begin{pmatrix} s + 2t \\ 2s + t \\ 3s \end{pmatrix}$$

for some real numbers s and t. In particular, the vector

$$\begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$$

is not in the span of the set $\{(1,2,3),(2,1,0)\}$. This implies that the set $\{(1,2,3),(2,1,0),(3,1,0)\}$ is linearly independent. Since it is a linearly independent set with three elements (which is the dimension of $\mathbb{R}^3$), it has to be a basis.

Q4. Prove: If $W \subseteq V$ is a subspace of a finite-dimensional vector space V then W is finite-dimensional.

Solution: Let α be a basis for W, and let β be a basis for V. Since α is a linearly independent subset of V, and β is a set of generators for V, we know that the amount of elements in α is at most the number of elements in β, but β is finite (!) so α has to be finite as well, and therefore W is finite-dimensional.

Q5. Suppose that V is finite-dimensional. Let W_1 and W_2 be subspaces of V with $V = W_1 + W_2$ and $W_1 \cap W_2 = \{\mathbf{0}\}$. Prove: $\dim(V) = \dim(W_1) + \dim(W_2)$.

Solution: Let $\alpha = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis for W_1 and $\beta = \{\mathbf{w}_1, \ldots, \mathbf{w}_k\}$ be a basis for W_2:

Claim 1: $\alpha \cup \beta$ is linearly independent.

Proof: Let

$$a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n + b_1\mathbf{w}_1 + \cdots + b_k\mathbf{w}_k = \mathbf{0}$$

be a null linear combination of the elements of $\alpha \cup \beta$. Then

$$a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = -b_1\mathbf{w}_1 - \cdots - b_k\mathbf{w}_k.$$

This implies that the vectors $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ and $\mathbf{w} = -b_1\mathbf{w}_1 - \cdots - b_k\mathbf{w}_k$ are elements of both W_1 and W_2. Since $W_1 \cap W_2 = \{\mathbf{0}\}$, we can conclude that $\mathbf{v} = \mathbf{w} = \mathbf{0}$. Now, since

$$a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$$

and α is linearly independent, $a_1 = \cdots = a_n = 0$. Also, since

$$-b_1\mathbf{w}_1 - \cdots - b_k\mathbf{w}_k = \mathbf{0}$$

and β is linearly independent, $b_1 = \cdots = b_k = 0$. Thus, $\alpha \cup \beta$ is linearly independent.

Claim 2: $\alpha \cup \beta$ is a basis for V.

Proof: We already proved that $\alpha \cup \beta$ is linearly independent, so it only remains to check that it is a spanning set for V. But this follows from the fact that $V = W_1 + W_2$ (!): if $\mathbf{u}$ is an arbitrary vector in V, let $\mathbf{v} \in W_1$ and $\mathbf{w} \in W_2$ be such that $\mathbf{u} = \mathbf{v} + \mathbf{w}$, and choose coefficients $a_1, \ldots, a_n, b_1, \ldots, b_k$ such that $\mathbf{v} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$ and $\mathbf{w} = b_1\mathbf{w}_1 + \cdots + b_k\mathbf{w}_k$. Then

$$\mathbf{u} = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n + b_1\mathbf{w}_1 + \cdots + b_k\mathbf{w}_k$$

and therefore $\mathbf{u} \in \text{span}(\alpha \cup \beta)$.

Finally, note that $\alpha \cap \beta = \varnothing$ (why?) so the number of elements in $\alpha \cup \beta$ equals the number of elements in α plus the number of elements in β. Since $\alpha \cup \beta$ is a basis for V, we can conclude that $\dim(V) = |\alpha \cup \beta| = |\alpha| + |\beta| = \dim(W_1) + \dim(W_2)$.

Q6. Find the dimension of the following subspaces.

(a) $W \subseteq \mathbb{R}^3$ given by $x + y + z = 0$.

Solution: By parametrizing $z = s$ and $y = t$, we have that every element of $\mathbb{R}^3$ satisfying the equation $x + y + z = 0$ is of the form

$$\begin{pmatrix} -s-t \\ t \\ s \end{pmatrix} = s\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + t\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

so the dimension of W equals 2.

(b) $W \subseteq \mathbf{P}_n(\mathbb{R})$ given by $W = \{f : f(a) = 0\}$.

Solution: If $f(a) = 0$ and $f(x) = b_0 + b_1 x + \cdots + b_n x^n$, then

$$b_0 + ab_1 + a^2 b_2 + \cdots + a^n b_n = 0$$

By parametrizing $b_i = t_i$ for each $1 \leq i \leq n$, we have that every polynomial $f(x) \in \mathbf{P}_n(\mathbf{R})$ satisfying that $f(a) = 0$ is of the form

$$f(x) = (-at_1 - a^2 t_2 - \cdots - a^n t_n) + t_1 x + \cdots + t_n x^n = t_1(x-a) + t_2(x^2 - a^2) + \ldots t_n(x^n - a^n)$$

so the dimension of W equals n.

(c) $W \subset \mathbb{R}^4$ given by the solutions of the following linear system

$$\begin{cases} x + y + z + w &= 0 \\ y + w &= 0 \\ z + w &= 0 \end{cases}$$

Solution: By parametrizing $w = t$, we have that every element of $\mathbb{R}^4$ satisfying the system above is of the form

$$\begin{pmatrix} t \\ -t \\ -t \\ t \end{pmatrix} = t\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

so the dimension of W equals 1.

Q7. Suppose that W_1 and W_2 are finite dimensional subspaces of V and

$$\dim(W_1) = m \leq n = \dim(W_2)$$

Prove:

(a) $\dim(W_1 \cap W_2) \leq n$

> **Solution:** Note that $W_1 \cap W_2$ is a subspace of W_2, so $\dim(W_1 \cap W_2) \leq \dim(W_2) = n$.

(b) $\dim(W_1 + W_2) \leq n + m$

> **Solution:** If α is a basis for W_1, and β is a basis for W_2, then their union $\alpha \cup \beta$ is a spanning set (maybe not a basis (!)) for $W_1 + W_2$ (*see the proof of* ***Claim 2*** *in Q5*). This means that any basis for $W_1 + W_2$ has at most as many elements as $\alpha \cup \beta$. Thus,
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> $$\dim(W_1 + W_2) \leq |\alpha \cup \beta| \leq |\alpha| + |\beta| = \dim(W_1) + \dim(W_2) = m + n.$$

Q8. Let $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \mathbf{w}\}$ be a subset of a vector space V. Consider the following subspaces:

$$W_1 = \operatorname{span}(\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}) \qquad W_2 = \operatorname{span}(\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \mathbf{w}\})$$

(a) Find a necessary and sufficient condition for $\dim(W_1) = \dim(W_2)$.

> **Solution:** Since $W_1 \leq W_2$, we have that $\dim(W_1) = \dim(W_2)$ if, and only if, $W_1 = W_2$ if, and only if, $\mathbf{w} \in W_1$, that is, if $\mathbf{w}$ was already a linear combination of the $\mathbf{v}_i$'s.

(b) State and prove a relationship between $\dim(W_1)$ and $\dim(W_2)$ when $\dim(W_1) \neq \dim(W_2)$.

> **Solution:** If $\dim(W_1) \neq \dim(W_2)$, then $\dim(W_2) \geq \dim(W_1) + 1$.
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> To see this, let $\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ be a basis for W_1 (note that m may be smaller than k). Since $\mathbf{w} \notin W_1$ (we are assuming that $\dim(W_1) \neq \dim(W_2)$), the collection $\{\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}\}$ is a linearly independent subset of W_2. Therefore,
>
> $$\dim(W_2) \geq |\{\mathbf{u}_1, \ldots, \mathbf{u}_m, \mathbf{w}\}| = |\{\mathbf{u}_1, \ldots, \mathbf{u}_m\}| + 1 = \dim(W_1) + 1.$$